

How well do sunken vessels approximate fish assemblages on coral reefs? Conservation implications of vessel-reef deployments

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Abstract The amount of artificial habitat (termed ‘artificial reef’, AR) in marine systems is rapidly increasing, yet the effect of most types of AR on reef communities remains unknown. We examined the role of well-established vessel-reefs in structuring coral reef fish assemblages by comparing assemblages on 7 World War II wrecks (>65 years old) to those on interspersed coral patch reefs of comparable size in a tropical lagoon. Fish abundance, species richness, diversity and feeding guild structure on wrecks were similar to natural reefs; however, species composition differed between the two reef types ($R = 0.189\text{--}0.341$, average dissimilarity: 67.3–68.8 %). Despite being more species-rich and diverse, fish assemblages on larger wrecks were less similar to assemblages on their adjacent natural reefs than smaller wrecks. Wrecks may also have affected fish abundance on adjacent natural reefs, with reefs adjacent to larger wrecks supporting higher abundances than reefs adjacent to smaller wrecks. Our results indicate that increases in vessel-reef habitat may not greatly affect reef fish assemblage parameters, but may affect the relative abundances of particular species.

Introduction

Coral reefs are declining worldwide (20 % loss, Hughes 1994; Gardner et al. 2003; Pandolfi et al. 2003; Wilkinson 2004),

and the amount of artificial reef (AR) habitat is increasing, through both the addition of marine infrastructure (e.g. breakwaters, oil rigs) and the deployment of specifically designed units for particular ecological purposes (e.g. fisheries enhancement reefs). ARs are now the dominant reef habitat in some areas (Chapman 2003), for example, the breakwaters constructed around the man-made Palm Islands off the coast of Dubai, United Arab Emirates, are 65 km long and represent more than a 50 % increase in hard-bottom habitat in the region (Burt et al. 2009). Despite this proliferation, little is known about the effects of increasing amounts of artificial habitat on marine communities. It has been suggested that ARs could help to conserve reef communities and protect against diversity loss by acting as surrogates for degraded natural reefs (e.g. Clark and Edwards 1994). In support, artificial reefs have been shown to develop similar abundances and species diversity to nearby natural reefs (Bohnsack et al. 1994; Clark and Edwards 1999; Rilov and Benayahu 2000; Perkol-Finkel and Benayahu 2004; Burt et al. 2009). However, results have varied widely among studies, and species composition has often been shown to differ between ARs and natural reefs (Clark and Edwards 1994; Clark and Edwards 1999; Perkol-Finkel et al. 2006; Burt et al. 2009). If ARs develop distinct communities, a substantial increase in AR habitat in an area could alter community structure away from a natural state and do little to maintain natural diversity levels. Determining how closely ARs can approximate communities on natural reefs is critical for an accurate assessment of their potential ecological benefit (or impact).

Most investigations have focused on the taxonomic similarity between AR and natural reef communities, and consequently, the ability of ARs to approximate other aspects of community structure and function on natural

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reefs remains poorly understood. Potential differences in the relative abundance of functional groups between ARs and natural reefs may be more important than differences in taxonomic composition, because changes in functional groups can affect ecosystem processes (Grime 1997; Bellwood et al. 2003; Mora et al. 2011). If communities on ARs are less functionally diverse than those on natural reefs, an increase in AR habitat may alter ecosystem function and reduce the resilience of reef communities to environmental perturbation (e.g. coral-algal phase shifts in the Caribbean, Hughes 1994; Bellwood et al. 2004). Few studies have compared the relative abundance of functional groups between ARs and natural reefs; however, differences in feeding guild structure have been identified for fish (Bohnsack et al. 1994; Arena et al. 2007). Determining whether differences in functional groups between ARs and natural reefs are common, and whether ARs can generally approximate functional diversity on natural reefs, is necessary for evaluating the relative quality of AR habitat.

In addition to developing different communities, ARs may actually affect the structure and function of communities on nearby natural reefs; however, this potential impact of ARs has rarely been investigated. Directly, ARs could affect nearby natural assemblages through the exchange of mobile taxa (e.g. fish attraction to ARs, Matthews 1985; Bohnsack 1989; Bohnsack et al. 1994). Indirect effects could include food-web alterations resulting from changes in the relative abundance of particular trophic groups (e.g. loss of herbivores to ARs), as well as changes in population-regulating processes, such as reduced recruitment (through recruit interception, e.g. Emery et al. 2006) and increased predation (export of predators from ARs). Only two studies have examined changes to natural fish assemblages following AR deployment (Stone et al. 1979; Masuda et al. 2010), and only one of these examined multiple ARs (Masuda et al. 2010). Further investigation of the potential of ARs to affect the assemblage structure and species composition of nearby natural reefs is essential for assessing any potential impacts of AR deployment.

Although specifically designed ARs are becoming more common, the use of ‘materials of opportunity’ (e.g. obsolete oil rigs, vessels) is still widespread. Sunken vessels (herein, ‘vessel-reefs’) are one type of AR which are becoming increasingly popular with both recreational users and environmental managers (Murray and Betz 1994; Arena et al. 2007). Vessel-reefs are highly regarded by recreational users for their ability to support high densities of fish, and the popularity of shipwrecks as fishing and dive sites has prompted the creation of new vessel-reefs through deployment of obsolete vessels. Despite their popularity, little is known about the fish assemblages that develop on vessel-reefs or how they compare to those on natural reefs.

It may be predicted that assemblages on vessel-reefs will differ substantially to those on natural reefs, due to considerable structural differences between the two reef types (Hixon and Beets 1993; Syms and Jones 2000). However, the few studies comparing fish assemblages between vessel-reefs and natural reefs have produced mixed results (Jones and Thompson 1978; Diamant et al. 1986; Arena et al. 2007; Honório et al. 2010). These studies have generally been hindered by the investigation of only a few vessel-reefs, little consideration of the effect of vessel-reef age on assemblage similarity (but see Arena et al. 2007) and incomplete interspersions of vessel-reefs and natural reefs. In addition, studies have rarely accounted for the effect of vessel-reef size on assemblage similarity to natural reefs and have usually compared isolated vessel-reefs with large continuous natural reefs. As deployment of vessel-reefs increases, there is a need for rigorous investigation of their ability to approximate the structure and function of fish assemblages on natural reefs. Such information will allow an assessment of their likely ecological benefit (or impact) and allow for informed decisions to be made regarding future deployments.

To examine whether vessel-reefs can approximate fish assemblages on surrounding natural reefs, we compared the structure and composition of fish assemblages associated with World War II ship and plane wrecks to coral patch reefs in a tropical lagoon. These wrecks provided a rare opportunity to examine fish assemblages on well-established (>65 years) vessel-reefs and compare them to assemblages associated with interspersed patch reefs of mostly comparable size. In addition to taxonomic structure, we compared feeding guild structure between wrecks and reefs to examine potential differences in functional groups between the two reef types. Because the wrecks ranged in size from a small seaplane through to a freighter ship and were all established at the same time (June, 1944), we were also able to examine the effect of vessel-reef size on assemblage structure, without the confounding effect of reef age. Finally, we compared paired wrecks and reefs to identify potential effects of vessel-reefs on the fish assemblages of nearby natural reefs.

Materials and Methods

Study area and site description

Wrecks and coral patch reefs (hereafter reefs) were located within Tanapag Lagoon on the north-western side of Saipan (15°10′51″ N, 145°45′21″ E), Northern Marianas Islands, in the central western Pacific Ocean. Saipan is the largest (115 km²) of the 15 islands and is located near the southern limit of the chain. The lagoon is 9 km long and

Table 1 Characteristics of 7 wrecks and adjacent coral patch reefs surveyed in Tanapag Lagoon on the north-western side of Saipan, Northern Marianas Islands

Wreck name	Wreck type	Wreck depth (m)	Vertical relief (m)	Wreck area (m ²)	Approx. area of adjacent reef (m ²)	Distance to natural reef (m)
Avenger	Plane	3	1.5	20	40	20
Jake	Plane	6	1.0	24	40	10
Mariner	Plane	7	2.0	50	50	10
Daihatsu	Ship	10	1.5	64	600	50
Emily	Plane	6	2.0	93	60	10
Subchaser	Ship	9	4.0	400	5000	30
Shoan	Ship	7	6.0	2000	40	15

1–3.5 km wide, and consists of low to medium relief (≤ 2 m) coral patch reefs interspersed with sand and rubble areas.

Seven wrecks (0.1–3.3 km apart) were surveyed within a 4.2 km² area of the lagoon. Wrecks included both planes and ships and were selected to provide a large size-range of artificial habitat (20–2,000 m², Table 1). All wrecks were sunk at the same time (June, 1944) and were located on bare sand in ≤ 10 m water depth (Table 1). Vertical relief of wrecks ranged from 1 to 6 m; however, only the *Subchaser* and *Shoan* wrecks exceeded 2 m relief (Table 1). Wreck structure consisted of flat horizontal and vertical surfaces punctuated by medium to large interstices (0.5–2 m diameter) and included numerous internal spaces and overhangs. Surfaces were covered with sparse turfing algae and supported few corals or other encrusting invertebrates compared to natural patch reefs (<10 % cover).

One natural patch reef was surveyed adjacent (≤ 50 m away) to each wreck (Table 1). Reefs were located in various compass directions relative to adjacent wrecks, resulting in interspersed wrecks and reefs within the study area. Reef size was estimated using a combination of underwater observations during surveys, surface observations from the research vessel, and measurements from satellite images (Google Earth, Google Incorporated). Reefs ranged in size from 40 to 5,000 m² and there was no relationship between reef size and size of adjacent wrecks (Table 1). Reef habitat consisted of emergent rock, supporting 50–90 % live cover of small- to medium-sized encrusting and massive corals. Reef surfaces were rugose, with numerous small (0.1–0.3 m diameter) holes, but few large holes and overhangs.

Survey design

To assess the consistency of any patterns in fish assemblage structure, wrecks and reefs were surveyed twice, once in mid-February and once in mid-June, 2010. Five wrecks and adjacent reefs were surveyed in February, and 7 wrecks (including the original 5) and adjacent reefs were surveyed in June. The smallest and largest wrecks were included in both the survey periods. Environmental

conditions were similar between sampling periods (current speeds ≤ 0.2 m s⁻¹, visibility ≥ 20 m); however, water temperatures were 25–26 °C in February and 28–29 °C in June.

Fishes were counted along 3 replicate belt transects (5 m $\times$ 2 m $\times$ 1.5 m) at each reef by a single experienced observer (AMF) using SCUBA. Longer transects could not be used because of the small size of some wrecks (Table 1). The size of transects was chosen to allow random allocation in space, and to reduce the chance of repeatedly surveying the same sections of habitat over the two sampling periods. Prior to the start of a transect, the abundances of larger more mobile species were recorded from a distance in order to minimise the effect of diver disturbance on species composition. Smaller species were then recorded while swimming along the transect at a constant speed of ~ 2 m min⁻¹. Fish were identified to species using Myers (1989) and Randall et al. (1997) and were also grouped into primary feeding guild, that is, grazer, scraper-excavator, planktivore, invertivore, coral-livore, piscivore-invertivore, and piscivore, using Froese and Pauly (2011).

Data analysis

Fish assemblage structure and composition were compared between wrecks and reefs using both multivariate and univariate techniques. The similarity of wreck and reef sites was examined using non-metric multidimensional scaling (nMDS) on ranked Bray-Curtis similarities calculated from multi-species abundance data. Separate analyses were carried out on abundance data at the species and feeding guild levels, as well as for the two sampling periods. In all cases, abundance data were square-root transformed to reduce the influence of the most abundant taxa. The significance of differences between groups was tested using one-way analysis of similarity (ANOSIM), and the species and feeding guilds contributing the most to differences between groups were identified using similarity percentage analysis (SIMPER; Clarke and Warwick 2001). Parrotfish (Scaridae) were excluded from the analyses at the species level, as this group could not be consistently

identified to species; however, parrotfish were included in analyses carried out at the feeding guild level.

Total fish abundance (number of individuals per 10 m²), species richness (S), and diversity (Shannon-Wiener, H') were compared between reef types (wrecks and reefs), sites (5 wrecks/reefs), and sampling periods (February and June) using 3-factor analyses of variance (ANOVAs). Reef type was treated as a fixed factor, while site and sampling period were treated as random factors, with site nested in reef type. In order to balance comparisons between sampling periods, the two extra wrecks (and adjacent reefs) surveyed in June were removed from these analyses. Total abundance and species richness data were $\log_{10}(x + 1)$ transformed to meet the assumptions of normality and homogeneity of variance. Diversity data did not require transformation.

Linear regression was used to test relationships between wreck size (area) and assemblage parameters on both wrecks and adjacent natural reefs. The relationships between wreck size and each of total fish abundance, species richness, and diversity were tested to determine whether larger wrecks supported more abundant, species-rich, and diverse fish assemblages than smaller wrecks. Relationships between wreck size and the total fish abundance, species richness, and diversity of adjacent natural reefs were tested to examine whether wrecks potentially affected fish assemblages on nearby natural reefs. The relationship between wreck size and Bray-Curtis similarities between wreck and reef pairs was used to determine whether assemblages on larger wrecks were more similar to adjacent natural reefs than smaller wrecks. Wrecks were paired with the nearest adjacent patch reef (Table 1). A p value <0.05 was considered significant for all tests.

Results

Assemblage structure and composition

We recorded 78 species in the study area; 67 on wrecks and 56 on reefs. Twenty-two species were found exclusively on wrecks (28 % of total species) and 11 were found exclusively on reefs (14 % of total species). Despite this, total abundance, species richness, and diversity did not differ significantly between wreck and reef habitats (all ANOVAs: $p > 0.14$). Mean ($\pm$ SE) species richness was higher in June (12.4 ± 0.56) than in February (8.3 ± 0.44 ; $F_{1,8} = 9669.75$, $p = 0.006$); however, abundance and diversity did not differ between the two sampling periods (abundance: $F_{1,8} = 14.69$, $p = 0.16$, diversity: $F_{1,8} = 104.73$, $p = 0.06$). Mean diversity varied significantly among sites within reef type ($F_{9,7} = 4.26$, $p = 0.03$).

Ordination revealed that the structure of fish assemblages differed between wrecks and reefs in both sampling periods (Fig. 1a, b). These differences were significant (ANOSIM: February, $R = 0.341$, $p < 0.001$; June, $R = 0.189$, $p < 0.001$); however, the low R -value in June indicated only minor differentiation of assemblages in this sampling period ($R < 2$, Clarke and Warwick 2001). SIMPER identified substantial dissimilarity in species composition between the two reef types in both sampling periods (average dissimilarity: February, 68.8 %; June, 67.3 %). Species contributing most to differences between reef types were similar between the two sampling periods (9 of 10 species in common, Table 2) and were primarily composed of pomacentrids (7 of 10 in February, 6 of 10 in June). Despite this, there was no clear preference for either wrecks or reefs (e.g. all species more abundant on wrecks) among pomacentrid species in terms of abundance (Table 2). Preference for reef type was consistent between the two sampling periods for most species; however, 3 species (*Dascyllus aruanus*, *Dascyllus reticulatus*, *Halichoeres trimaculatus*) were more abundant on one reef type in one sampling period and then more abundant on the other reef type in the other sampling period.

In contrast to the taxonomic structure of assemblages, feeding guild structure did not differ between wrecks and reefs in February (Fig. 1c, ANOSIM: $R = 0.07$, $p = 0.07$), and although a significant difference was found between wrecks and reefs in June (Fig. 1d, ANOSIM: $R = 0.08$, $p = 0.02$), the R -value was negligible. Guild diversity was similar between wrecks and reefs, with all 7 feeding guilds found on wrecks and 6 guilds found on natural reefs (piscivores were absent on this reef type).

Effect of wreck size on assemblage structure

Strong positive relationships were found between wreck size (area) and both mean species richness (Fig. 2a; $r^2 = 0.60$, $df = 6$, $p = 0.04$) and mean diversity (Fig. 2b; $r^2 = 0.93$, $df = 6$, $p < 0.001$) in June, but not in February (species richness: $r^2 = 0.29$, $df = 4$, $p = 0.35$; diversity: $r^2 = 0.32$, $df = 4$, $p = 0.32$). Mean total abundance appeared to be positively related to wreck size in February; however, the result was not significant ($r^2 = 0.69$, $df = 4$, $p = 0.08$). The relationship was also not significant in June ($r^2 = 0.13$, $df = 6$, $p = 0.43$). The relationship between wreck size and Bray-Curtis similarity between wreck and reef pairs was examined to determine whether assemblages on larger wrecks were more similar to natural reefs than smaller wrecks. A significant negative relationship between wreck size and Bray-Curtis similarity between wreck and reef pairs was found in June (Fig. 3; $r^2 = 0.64$, $df = 6$, $p = 0.03$), but not in February ($r^2 = 0.07$, $df = 4$, $p = 0.66$).

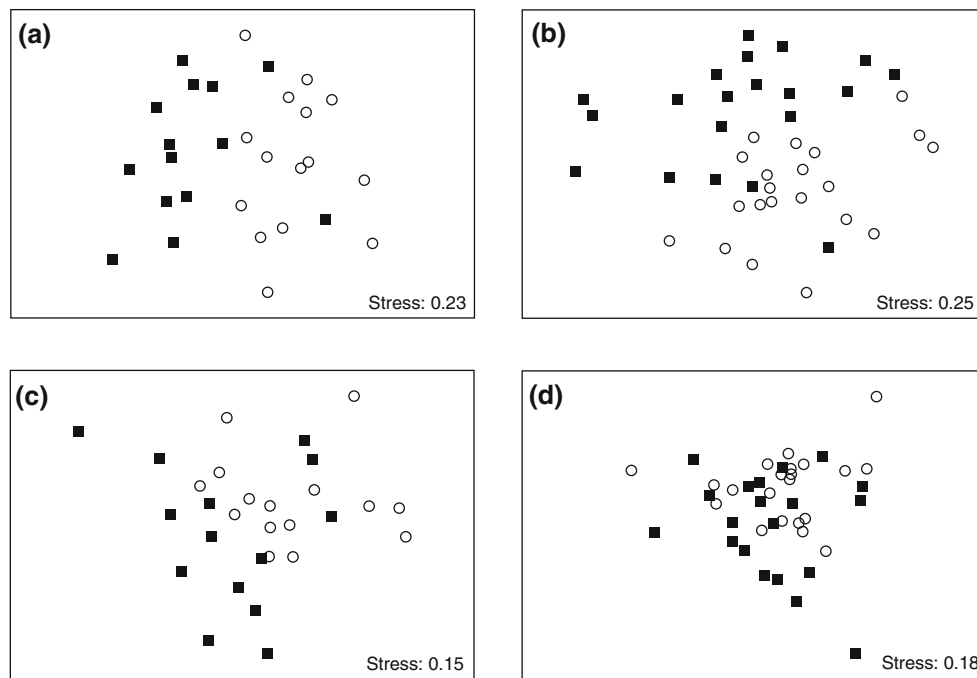


Fig. 1 Ordination results of multivariate fish assemblage comparisons between wreck (*squares*) and reef (*circles*) sites for species data in **a** February and **b** June, and feeding guild data in **c** February and **d** June 2010 in Tanapag Lagoon, Saipan

Effect of adjacent wrecks on reefs

A strong positive relationship was found between mean fish abundance on reefs and size of the adjacent wreck in June (Fig. 4a; $r^2 = 0.63$, $df = 6$, $p = 0.03$), but not in February ($r^2 = 0.50$, $df = 4$, $p = 0.18$). A positive relationship was also found between mean species richness and size of the adjacent wreck in February (Fig. 4b; $r^2 = 0.84$, $df = 4$, $p = 0.03$), but not in June ($r^2 = 0.08$, $df = 6$, $p = 0.55$). No significant relationships were found between mean diversity and size of the adjacent wreck in either sampling period (February: $r^2 = 0.10$, $df = 4$, $p = 0.61$; June: $r^2 = 0.18$, $df = 6$, $p = 0.34$). To determine whether assemblages on reefs were more similar to their adjacent wreck than to the other reefs, the Bray-Curtis similarity between each wreck and adjacent reef was compared to similarities between the reef and each other reef. In February, assemblages on all reefs were more similar to at least 3 out of 4 other reefs than their adjacent wreck. In June, assemblages on all reefs were more similar to at least 4 out of 6 other reefs than their adjacent wreck. The exception was the assemblage on the reef adjacent to the Avenger wreck, which was more similar to the wreck assemblage than any other reef.

Discussion

Mixed results among previous studies and limitations of experimental design have made it difficult to assess

whether vessel-reefs are generally capable of approximating fish assemblages on natural reefs. Fish abundances, species richness and diversity levels on vessel-reefs have been found to be similar or different to those on surrounding natural reefs, depending on the study considered (Jones and Thompson 1978; Diamant et al. 1986; Arena et al. 2007; Honório et al. 2010). Results from most studies are also difficult to interpret, because typically only 1–2 vessel-reefs were investigated, or the potential confounding effects of vessel-reef age and size were not considered, or both. In the most comprehensive investigation to date, Arena et al. (2007) compared fish assemblages on 6 vessel-reefs to those on surrounding coral reefs in Florida and found that both fish abundances and species richness differed between the two reef types. However, vessel-reefs were orders of magnitude smaller than surrounding coral reefs, and the two reef types were not completely interspersed. Assemblage comparisons may therefore have been partially confounded by size- or location-dependent assemblage development (Carr and Hixon 1997). Arena et al. (2007) also identified a trend of increasing species richness with increasing vessel-reef age, suggesting assemblages may not yet have stabilised on some of their vessel-reefs (3 out of 6 reefs <10 years old, Relini et al. 2002; Thanner et al. 2006). Because the wrecks in the current study were very old (>65 years), and of mostly comparable size to surrounding coral patch reefs, they provided a good test of the effect of vessel-reef habitat on fish assemblages, while minimising potential confounding

Table 2 Mean abundance and percentage contribution of the top 10 fish species contributing to assemblage differences between wrecks and reefs in both sampling periods, as indicated by SIMPER analysis

Species	Feeding guild	Wreck mean abundance (per 10 m ²)	Reef mean abundance (per 10 m ²)	Contribution (%)	Cumulative contribution (%)
February					
<i>Pomacentrus vaiuli</i>	planktivore	2.07	10.07	11.31	11.31
<i>Chromis viridis</i>	planktivore	2.50	8.20	8.81	20.11
<i>Dascyllus aruanus</i>	planktivore	2.50	2.13	6.82	26.93
<i>Abudefduf sexfasciatus</i>	planktivore	2.57	1.07	6.63	33.57
<i>Stegastes nigricans</i>	grazer	2.79	0.87	6.36	39.93
<i>Dascyllus reticulatus</i>	planktivore	1.36	2.33	5.74	45.67
<i>Acanthurus nigricans</i>	grazer	1.50	0.33	4.88	50.55
<i>Ctenochaetus striatus</i>	grazer	2.36	1.93	4.26	54.81
<i>Halichoeres trimaculatus</i>	invertivore	0.14	0.87	3.19	58.00
<i>Dascyllus trimaculatus</i>	planktivore	0.64	0.0	3.06	61.06
June					
<i>Pomacentrus vaiuli</i>	planktivore	2.65	12.33	7.87	7.87
<i>Dascyllus aruanus</i>	planktivore	4.10	4.38	6.74	14.61
<i>Stegastes nigricans</i>	grazer	3.10	1.43	5.26	19.87
<i>Chromis viridis</i>	planktivore	2.65	9.43	4.56	24.43
<i>Dascyllus reticulatus</i>	planktivore	2.25	2.24	4.56	28.99
<i>Chaetodon citrinellus</i>	invertivore	4.65	0.19	4.20	33.19
<i>Acanthurus nigricans</i>	grazer	1.55	1.00	3.54	36.73
<i>Abudefduf sexfasciatus</i>	planktivore	1.30	0.81	3.50	40.23
<i>Halichoeres trimaculatus</i>	invertivore	1.00	0.95	3.10	43.33
<i>Ctenochaetus striatus</i>	grazer	2.85	2.38	3.04	46.36

effects. Our results support the conclusion that well-established vessel-reefs are capable of approximating fish abundances and assemblage parameters on natural coral reefs.

Despite similarity in assemblage parameters, species composition was found to differ between wrecks and surrounding coral reefs. Although multivariate differences were not large ($R < 0.4$, Clarke and Warwick 2001), twenty per cent (11 of 56) of species found on reefs were not found on wrecks, and wrecks supported an additional 22 species not found on reefs. Differences in species composition were also generally consistent between the two sampling periods; of the top 10 species contributing to assemblage differences between wrecks and reefs, 9 were the same between sampling periods, and most species were consistently more abundant on one reef type (Table 2). This consistency indicates that observed differences in species composition between wrecks and reefs are unlikely to represent an anomalous event; however, further sampling is required to determine the degree of both intra- and inter-annual variability in compositional differences between the two reef types. Previous studies comparing species composition of fish assemblages between vessel-reefs and coral reefs have produced mixed results. Arena et al. (2007) found strong differences in composition between

the two reef types (74 % dissimilarity) on the Florida coast, whereas Honório et al. (2010) found little difference in composition between the two reef types on the Brazilian coast. Given the current study found moderate differences in species composition, it is clear that further research is needed to confirm both the generality and extent of compositional differences between vessel-reefs and coral reefs. Differences in species composition between other types of ARs and coral reefs have often been reported (Bohnsack et al. 1994; Rooker et al. 1997; Clark and Edwards 1999; Rilov and Benayahu 2000; Burt et al. 2009) and similar to the current study, have been found to occur despite similar levels of diversity between the two reef types (Rooker et al. 1997; Burt et al. 2009). If this pattern is generally confirmed for vessel-reefs, large additions of vessel-reef habitat may not overly affect fish diversity within an area, but may affect relative abundances of particular species.

The differences in species composition between wrecks and reefs in the current study were most likely caused by differences in habitat structure and composition, because other major differences (i.e. reef age, size, relative position) between the two reef types were either minimised or eliminated. Wrecks and reefs differed markedly in both physical and biological habitat characteristics; wrecks had

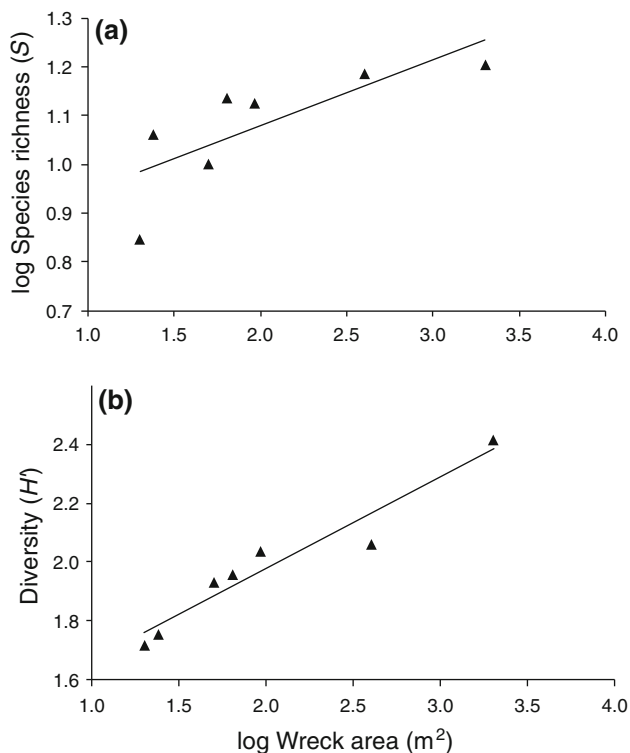


Fig. 2 Relationships between wreck size and **a** fish species richness and **b** diversity, in June 2010 in Tanapag Lagoon, Saipan. Note, axes do not start at 0

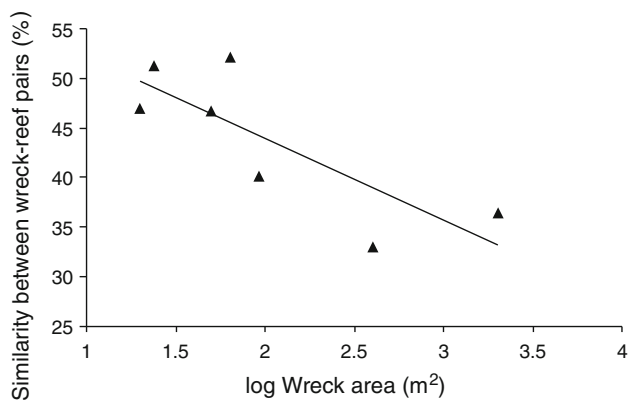


Fig. 3 Relationship between wreck size and the Bray-Curtis similarity index between paired wreck and reef sites in June 2010 in Tanapag Lagoon, Saipan. Note, axes do not start at 0

larger interstices and more vertical surfaces, but lower rugosity and coral cover, than natural reefs (see [Materials and Methods](#)). Such habitat characteristics are known to influence species composition, due to species-specific microhabitat preferences (Syms and Jones 2000) and the effect of refuge size and availability on levels of predation and competition (Shulman 1984; Hixon and Beets 1993). Given the old age of wrecks (>65 years), results here

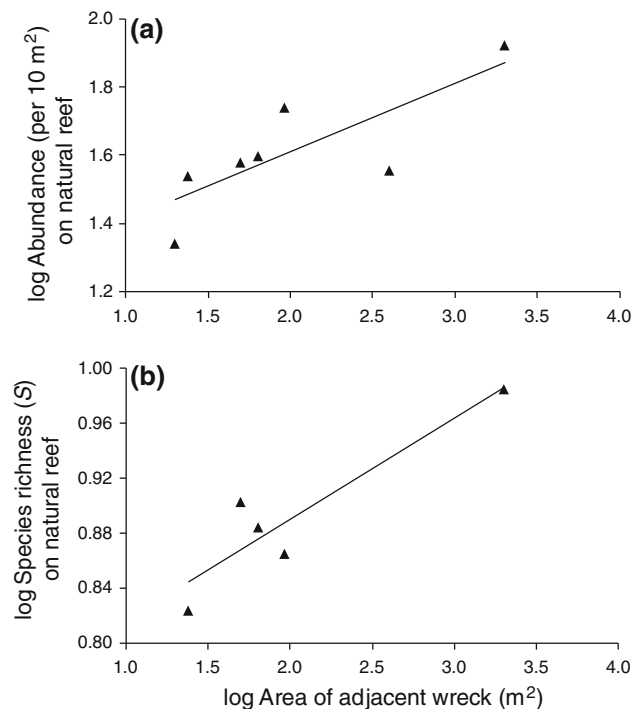


Fig. 4 Relationships between size of the adjacent wreck and **a** fish abundance on natural reefs in June and **b** species richness on natural reefs in February 2010 in Tanapag Lagoon, Saipan. Note, axes do not start at 0

indicate that the differences in species composition between vessel-reefs and natural reefs are likely to persist while habitat structure and composition remain different, that is, prolonged establishment may not overcome the effect of underlying differences in habitat. This has broader implications for the use of ‘materials of opportunity’ (e.g. car bodies) for planned ARs, as well as the installation of marine infrastructure, both of which rarely approximate the habitat characteristics of natural reefs. However, while different, the habitat provided by wrecks in the current study did not appear to be generally deficient relative to natural reef habitat. There was no clear preference for either reef type among the main species responsible for assemblage differences between wrecks and reefs (abundances, Table 2), as might be expected if one reef type provided generally superior resources (e.g. food, shelter) compared to the other. Therefore, even AR habitat that differs considerably to natural reef habitat may have conservation value, particularly in circumstances where natural reefs are in severe decline and closely approximating assemblages is not a priority (Clark and Edwards 1999).

Most investigations of fish assemblages on ARs have focused on taxonomic structure, and consequently, little is known about the ability of ARs to approximate other aspects of assemblage structure and function on natural reefs. We examined the similarity of feeding guilds between wrecks and natural reefs and found that all feeding

guilds present on natural reefs were present on wrecks. Further, multivariate analysis revealed only minor differences in guild structure between the two reef types ($R = 0.08$). These results suggest that the wrecks may function similarly to surrounding coral reefs in regard to trophic interactions, but comparisons of fish diets and feeding rates between the two reef types would be required to confirm this. Our results contrast with those of most previous studies, which have found differences in particular feeding guilds between ARs and natural reefs; most commonly, a higher abundance of planktivores on ARs (Bohnsack et al. 1994; Rilov and Benayahu 2000; Arena et al. 2007). Unlike the current study, vertical relief of ARs in these studies was generally greater than natural reefs, therefore higher abundances of planktivores on ARs may have resulted from increased feeding opportunities higher up in the water column (Rilov and Benayahu 2000; Arena et al. 2007). However, ARs were also not completely interspersed with natural reefs in these studies and differences in feeding guilds may have simply resulted from spatial variation in available food resources (e.g. more plankton in areas with strong current, Arena et al. 2007), rather than differences in foraging opportunities resulting from habitat differences. Our results indicate that vessel-reefs are capable of approximating feeding guild structure on coral reefs, but this may depend on the similarity of particular habitat characteristics between the two reef types, such as vertical relief. Further investigation of the effect of particular habitat characteristics on the relative abundance of functional groups would assist in determining which characteristics of vessel-reefs, and ARs in general, are most important for approximating assemblage function on natural reefs.

Understanding the effects of AR size on fish assemblages is important for evaluating whether particular AR installations (planned or otherwise) are likely to approximate assemblages on nearby natural reefs. For example, knowledge of the minimum size of AR required to achieve a specific level of diversity may prevent the installation of ARs that are too small to reach that level. Fish abundances and species richness have previously been found to increase with the size of specifically designed AR units (Bohnsack et al. 1994; Jan et al. 2003; Jordan et al. 2005); however, the current study is the first to investigate potential trends in the structure and composition of fish assemblages with the size of vessel-reefs. The wrecks examined were useful for this purpose, because they ranged greatly in size (20–2,000 m²) and were all the same age, thereby eliminating potential confounding effects of age-dependent assemblage development. Strong positive relationships were found between wreck size and both species richness and diversity in the June sampling period. These results indicate that larger vessel-reefs may develop

more diverse fish assemblages than smaller vessel-reefs and are similar to positive relationships found between reef size and community measures on natural reefs (Gladfelter et al. 1980; Chittaro 2002). The absence of relationships between wreck size and assemblage parameters in February may have resulted from low power due to the fewer wrecks (5 compared to 7) surveyed in this sampling period. Alternatively, relationships between wreck size and assemblage parameters may be temporally variable and further investigation at a range of temporal scales is required to confirm patterns observed here.

Although larger ARs may develop more species-rich and diverse fish assemblages than smaller ARs, this does not ensure that they will better approximate assemblage structure and composition on nearby natural reefs. To our knowledge, no previous study has investigated the relationship between AR size and the similarity of fish assemblages on paired AR and natural reef sites. We found a negative relationship between wreck size and the Bray-Curtis similarity between each wreck and its adjacent patch reef in the June sampling period. This indicates that although larger wrecks had greater species richness and diversity in this sampling period, their assemblages were less similar to adjacent natural reefs than smaller wrecks. Smaller wrecks generally matched the size of their adjacent reefs more closely than larger wrecks (Table 1). Therefore, the pattern observed here may have resulted from size-dependent assemblage development coupled with greater size-differences between larger wrecks and their adjacent reefs. Further research is required to confirm the effect of differences in reef size on the ability of vessel-reefs, and other types of ARs, to approximate assemblages on nearby natural reefs. If our results reflect generally occurring patterns, ARs that differ greatly in size to surrounding natural reefs may be less likely to approximate natural assemblages. Such deployments should be considered carefully, particularly when they are to occur in close proximity to natural reefs.

The deployment of large ARs could alter the assemblage structure and function of nearby natural reefs; however, there has been minimal investigation of this potentially negative impact. Matthews (1985) documented movement of numerous fish species from natural reefs to a new AR up to 1.6 km away, yet no movement of fish from the AR to natural reefs. This indicated that ARs have the potential to affect assemblage structure on natural reefs by attracting more mobile species away from natural reefs; however, differences in assemblages between the two reef types were not found (Matthews 1985). We found that assemblages on reefs were generally less similar to their adjacent wreck than most other reefs in both sampling periods, indicating that any effect of the adjacent wreck on natural reef assemblages was less than the effect of habitat differences

between the two reef types. The exception to this pattern was the Avenger wreck and adjacent reef, the assemblages of which were more similar to each other than any other reef. This likely resulted from the shallow depth (3 m) of this site compared to other wrecks (6–10 m depth), and possibly because this site was furthest away from all other sites (1.4 km). We also investigated the relationships between each of fish abundance, species richness and diversity on natural reefs and the size of adjacent wrecks and found that both fish abundance and species richness on reefs were positively related to the size of adjacent wrecks in at least one sampling period. These relationships suggest that wrecks influenced neighbouring reef assemblages to some degree, because there was no relationship between the size of reefs and their adjacent wrecks (i.e. larger reefs were not simply paired with larger wrecks). However, a causal relationship cannot be confirmed here, because reef assemblages were not surveyed both before and after the addition of wrecks. Additionally, the relationship between species richness on reefs and size of the adjacent wreck in February is primarily driven by the smallest and largest data points, and a similar relationship was not detected in June. Our results suggest that deployment of large vessel-reefs in close proximity to natural reefs may affect natural fish assemblages; however, effects may be subtle, because reef assemblages were still more similar to each other than assemblages on adjacent wrecks. Effects may also be temporally variable, because most results were not consistent between the two sampling periods. Ultimately, manipulative experiments in which natural reef assemblages are surveyed both before and after the addition of vessel-reefs are required to confirm whether vessel-reefs are generally capable of influencing existing reef fish assemblages. Such experiments would also be valuable for testing whether other types of ARs are capable of affecting fish assemblages on nearby natural reefs.

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